

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

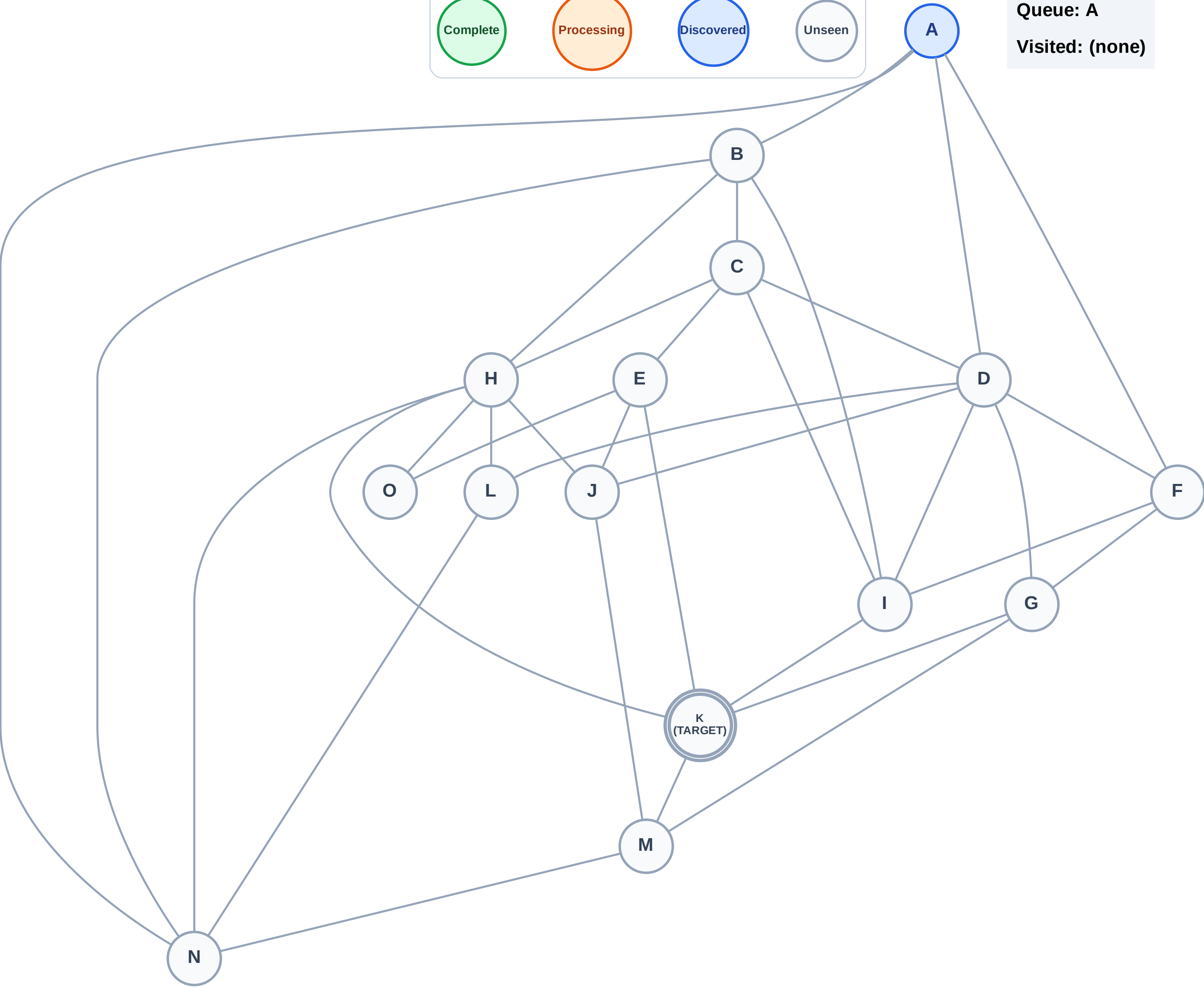
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



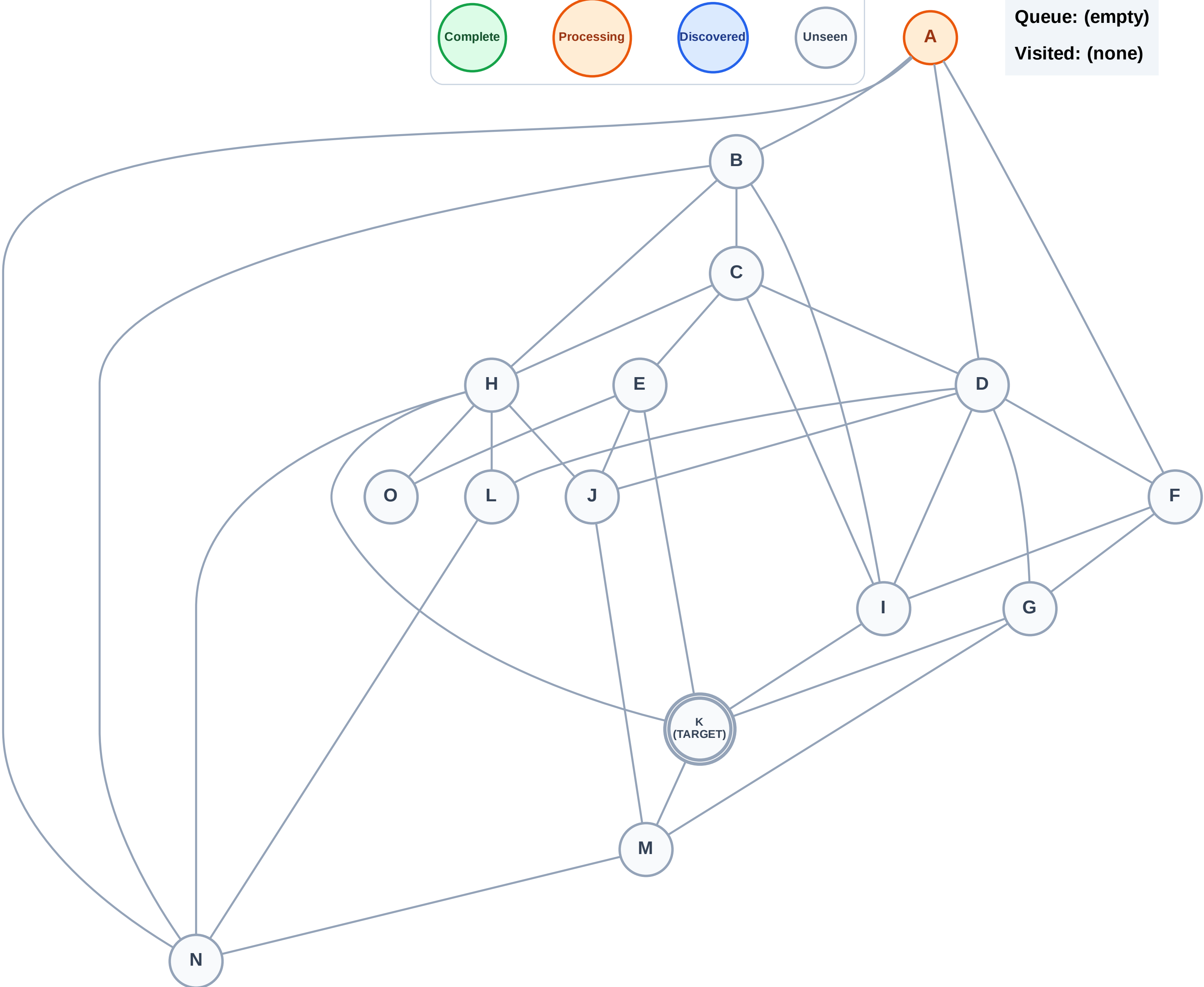
BFS Traversal

Step 2: Process node A

Legend



Queue: (empty)
Visited: (none)

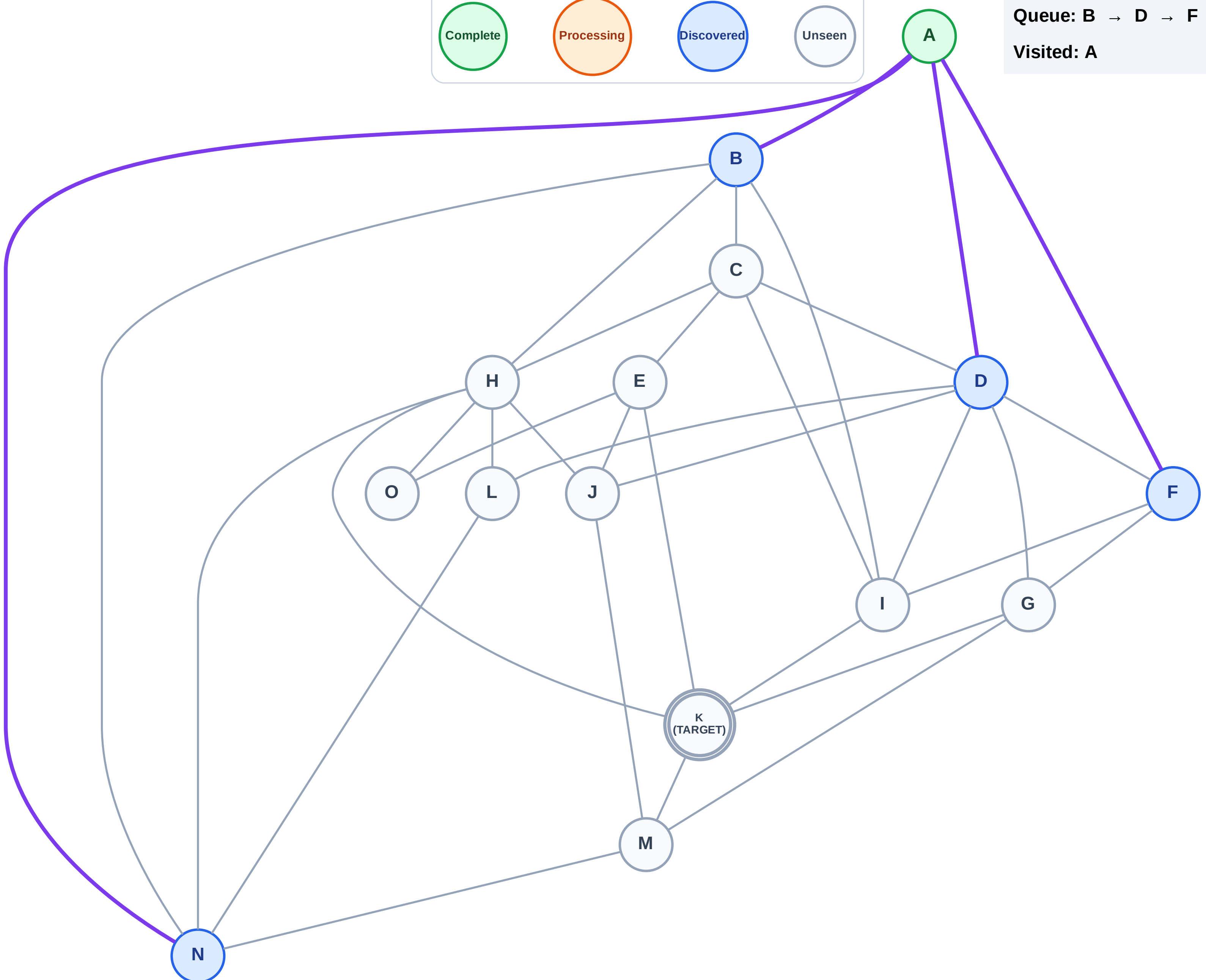


Step 3: Discover B, D, F, N from A

Legend

Unseen

Visited: A



BFS Traversal

Step 4: Process node B

Legend

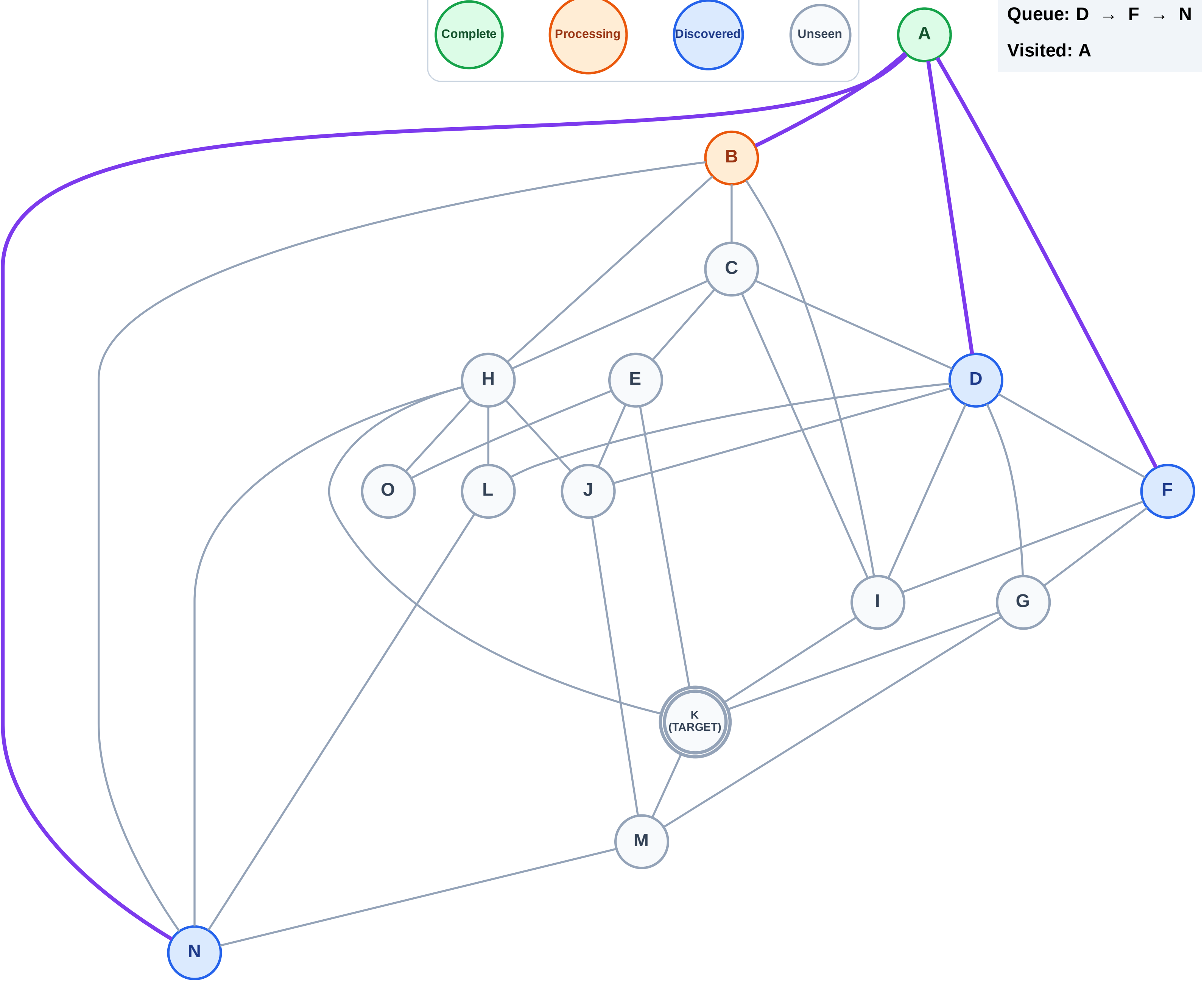
Complete

Processing

Discovered

Unseen

Queue: D → F → N
Visited: A



BFS Traversal

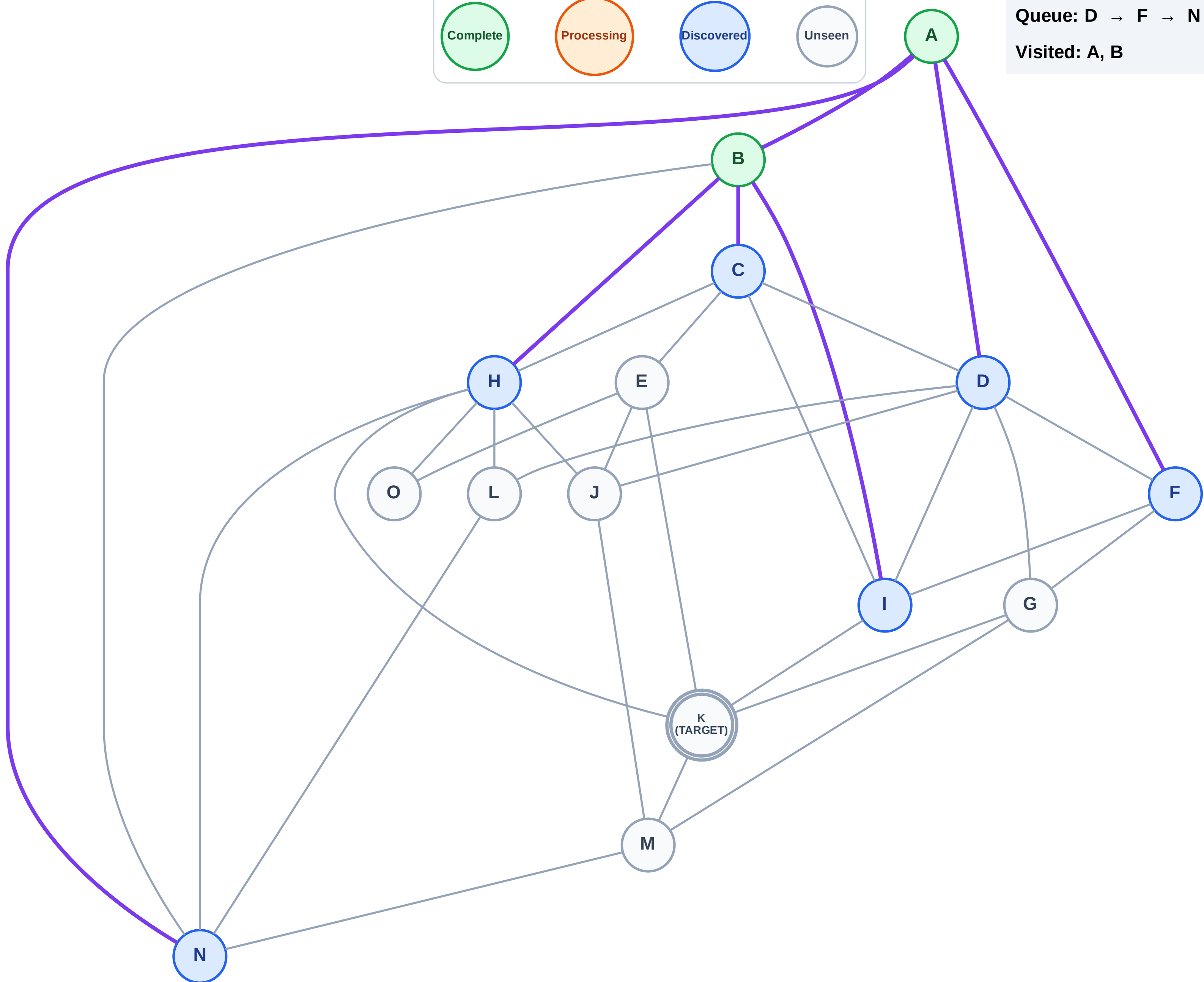
Step 5: Discover C, H, I from B

Legend



Queue: D → F → N → C → H → I

Visited: A, B



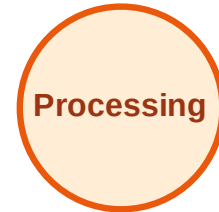
BFS Traversal

Step 6: Process node D

Legend



Complete



Processing



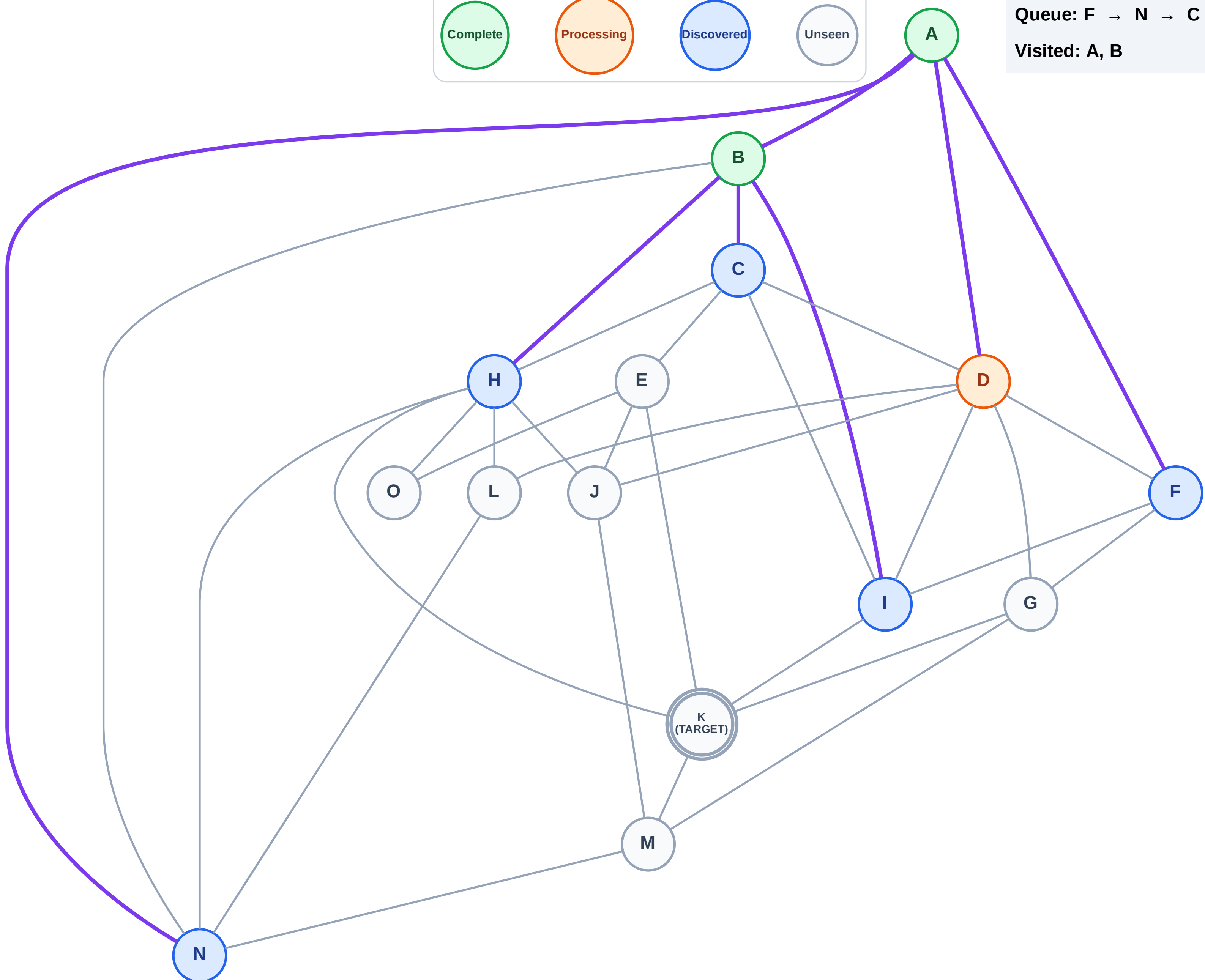
Discovered



Unseen

Queue: F → N → C → H → I

Visited: A, B



BFS Traversal

Step 7: Discover G, J, L from D

Legend



Complete



Processing



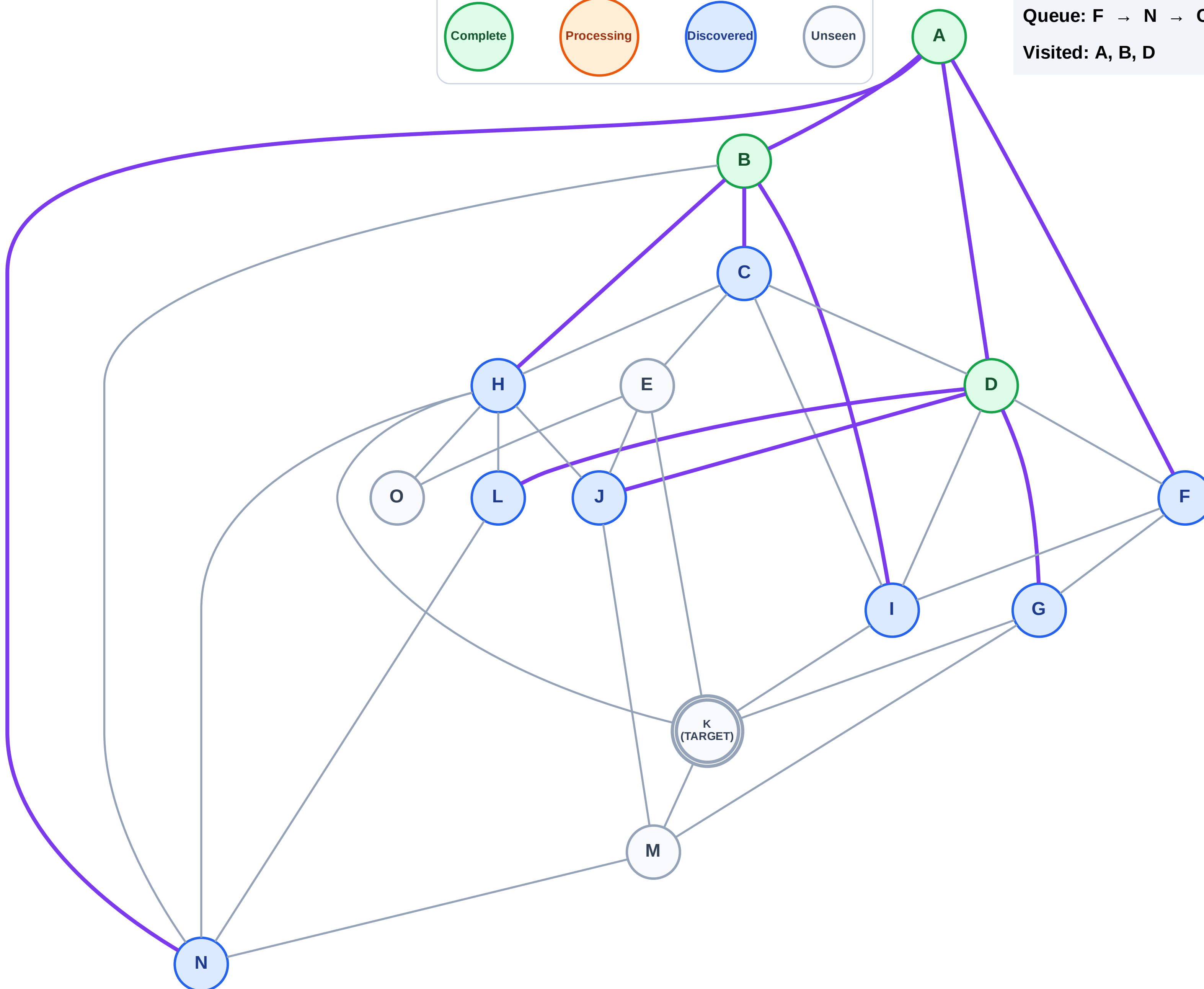
Discovered



Unseen

Queue: F → N → C → H → I → G → J → L

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

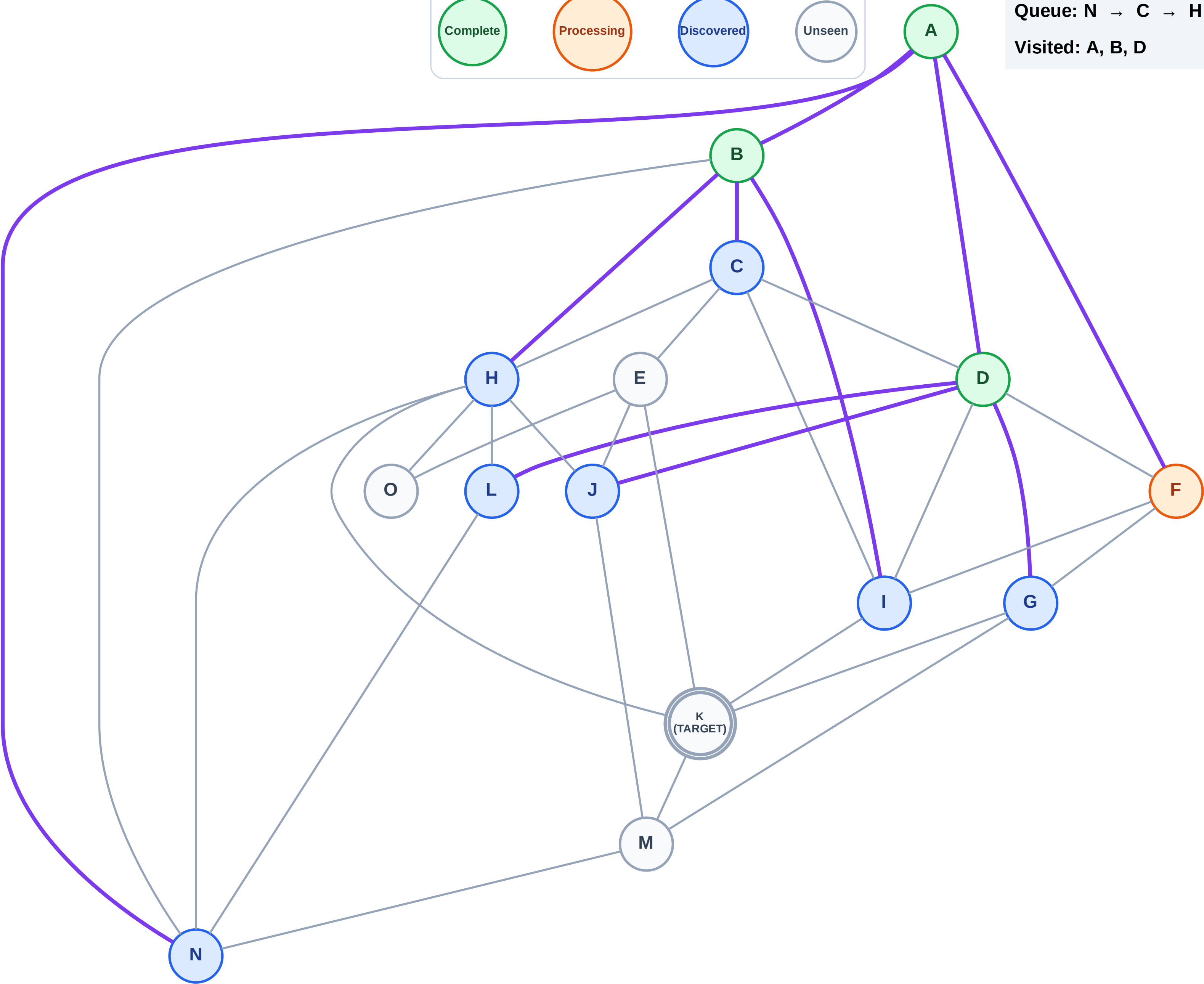
Processing

Discovered

Unseen

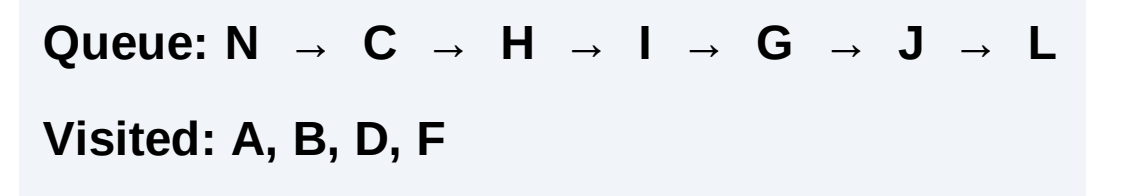
Queue: N → C → H → I → G → J → L

Visited: A, B, D



Step 9: No new neighbors from F

Legend



BFS Traversal

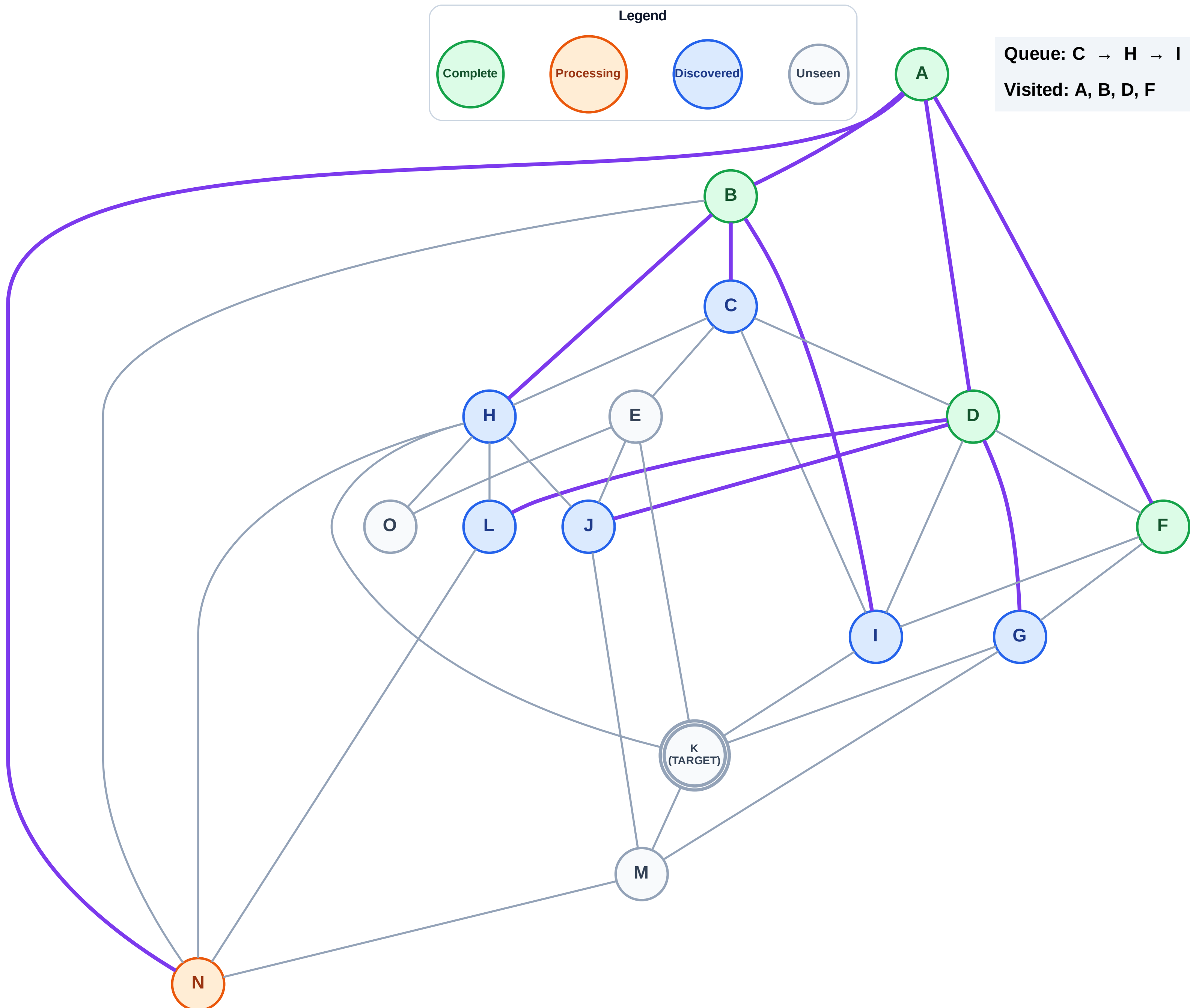
Step 10: Process node N

Legend



Queue: C → H → I → G → J → L

Visited: A, B, D, F



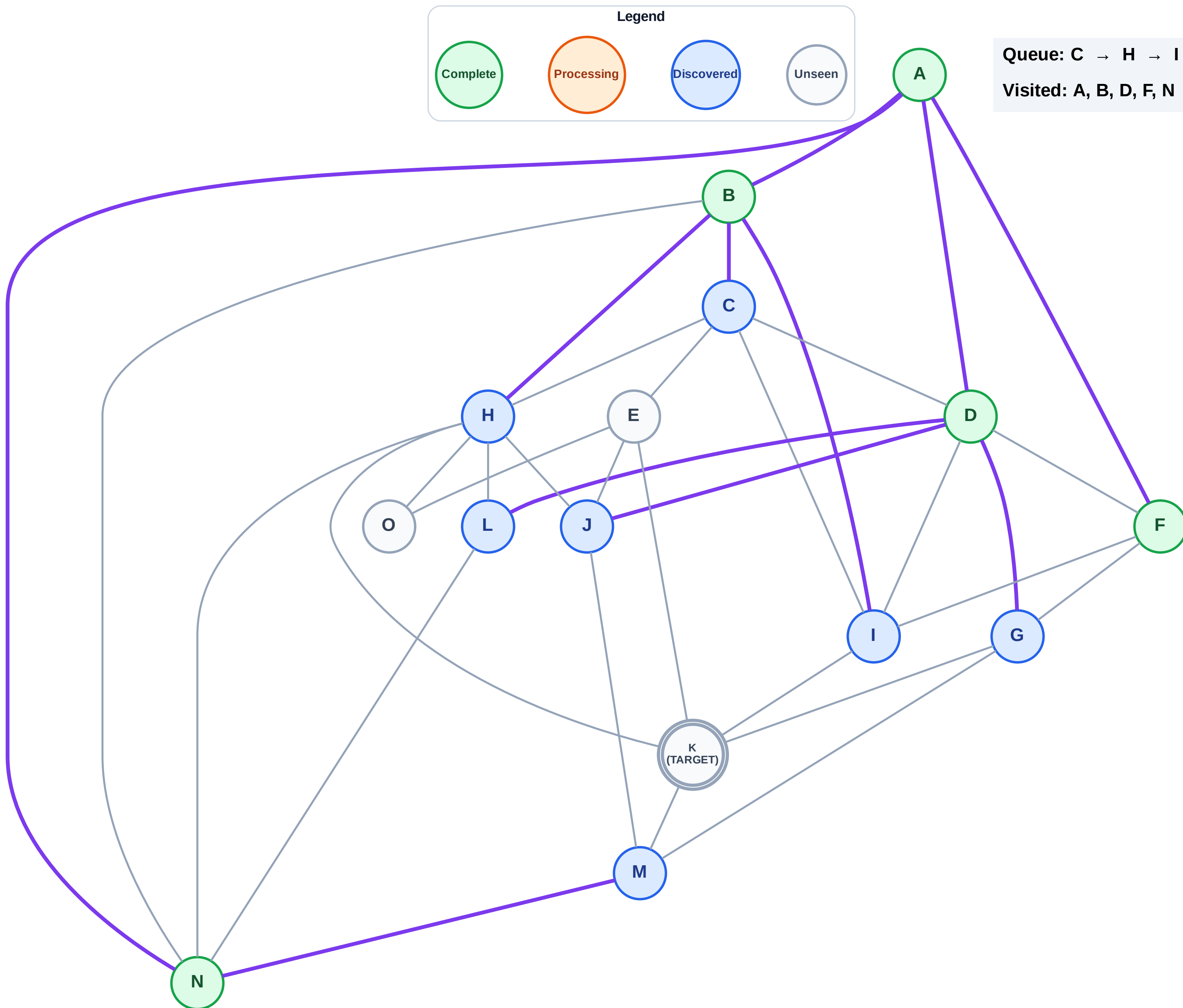
BFS Traversal

Step 11: Discover M from N

Legend



Queue: C → H → I → G → J → L → M
Visited: A, B, D, F, N



BFS Traversal

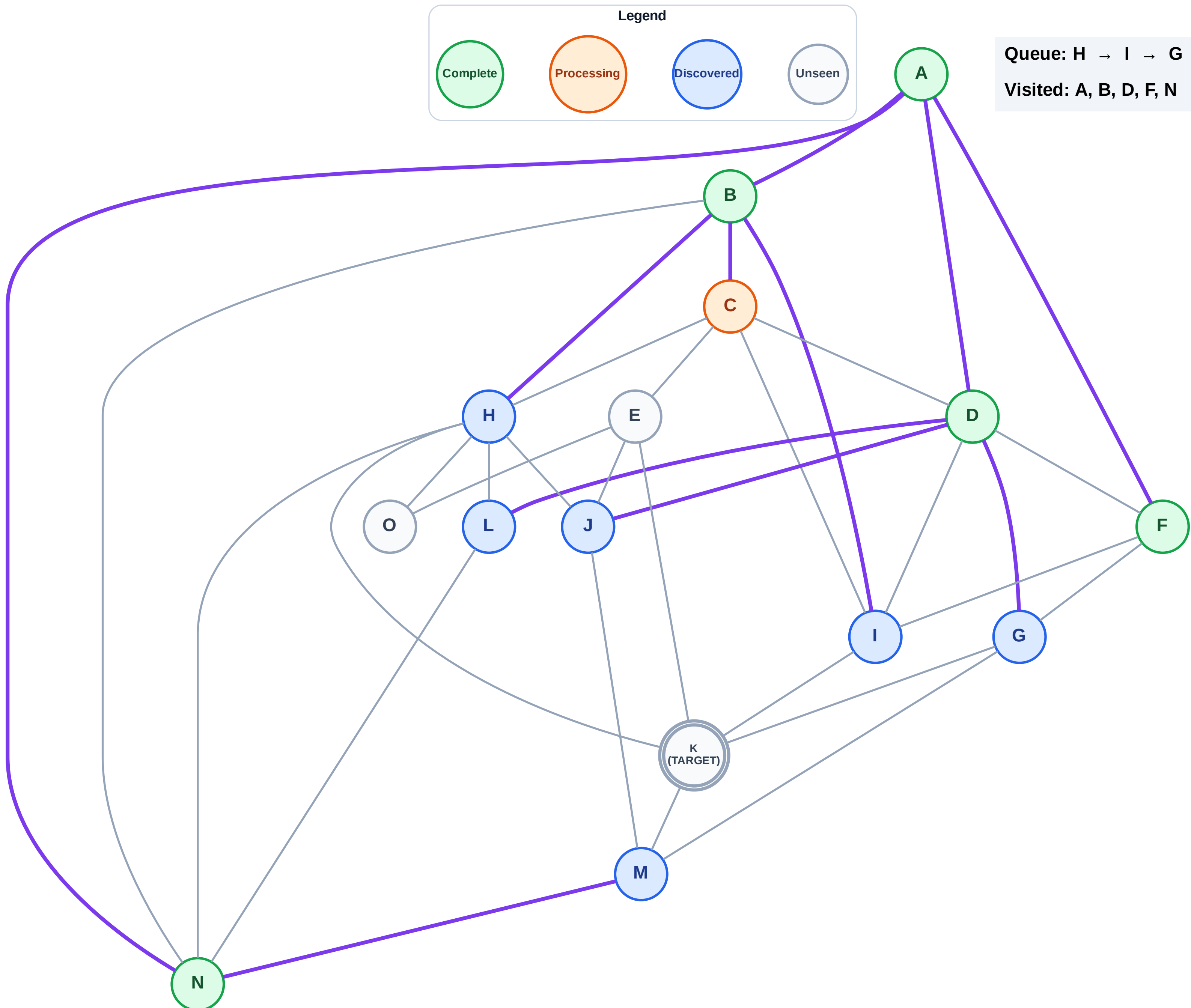
Step 12: Process node C

Legend



Queue: H → I → G → J → L → M

Visited: A, B, D, F, N



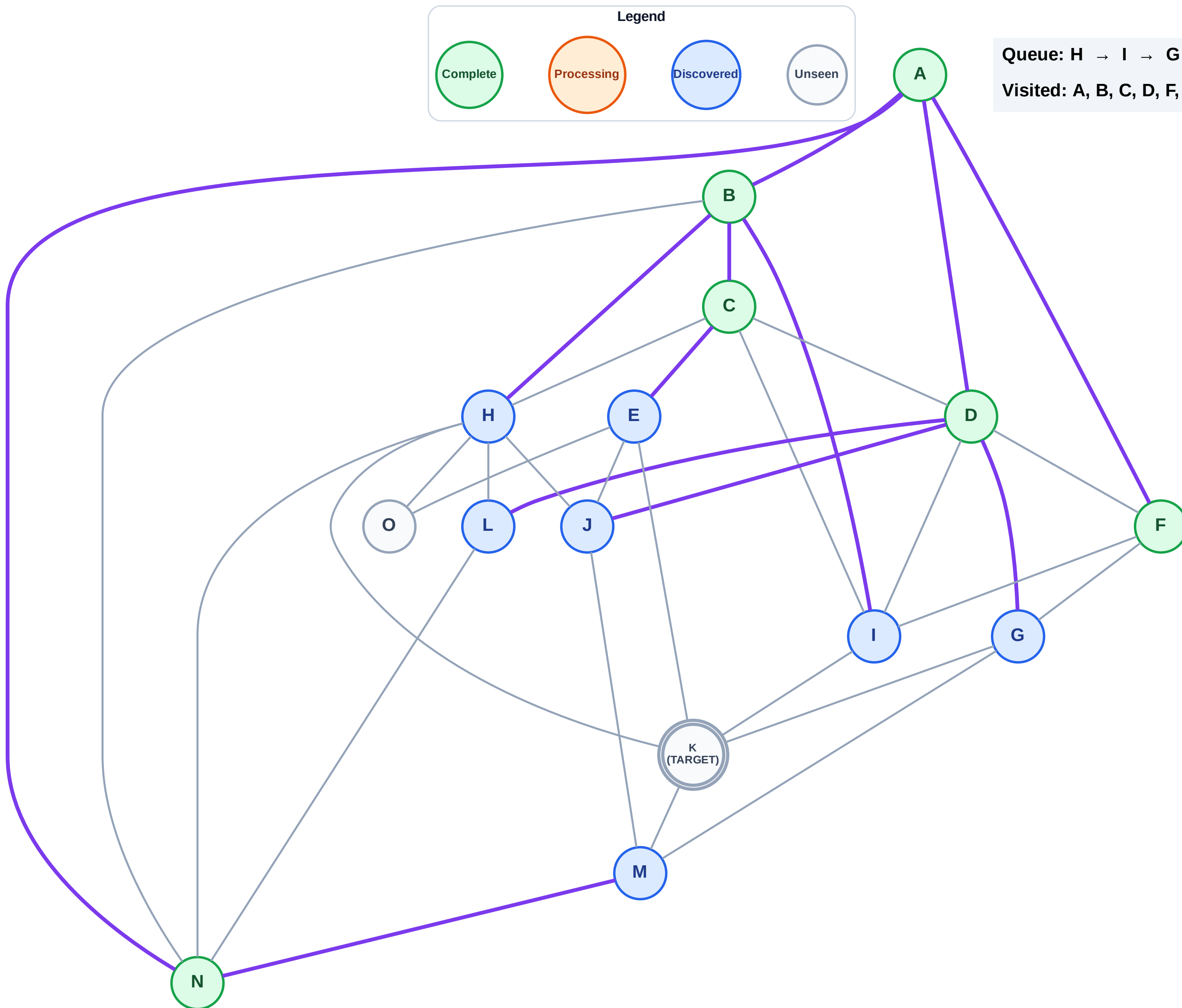
BFS Traversal

Step 13: Discover E from C

Legend



Queue: H → I → G → J → L → M → E
Visited: A, B, C, D, F, N



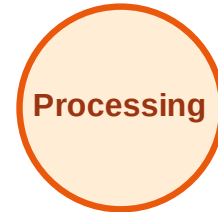
BFS Traversal

Step 14: Process node H

Legend



Complete



Processing



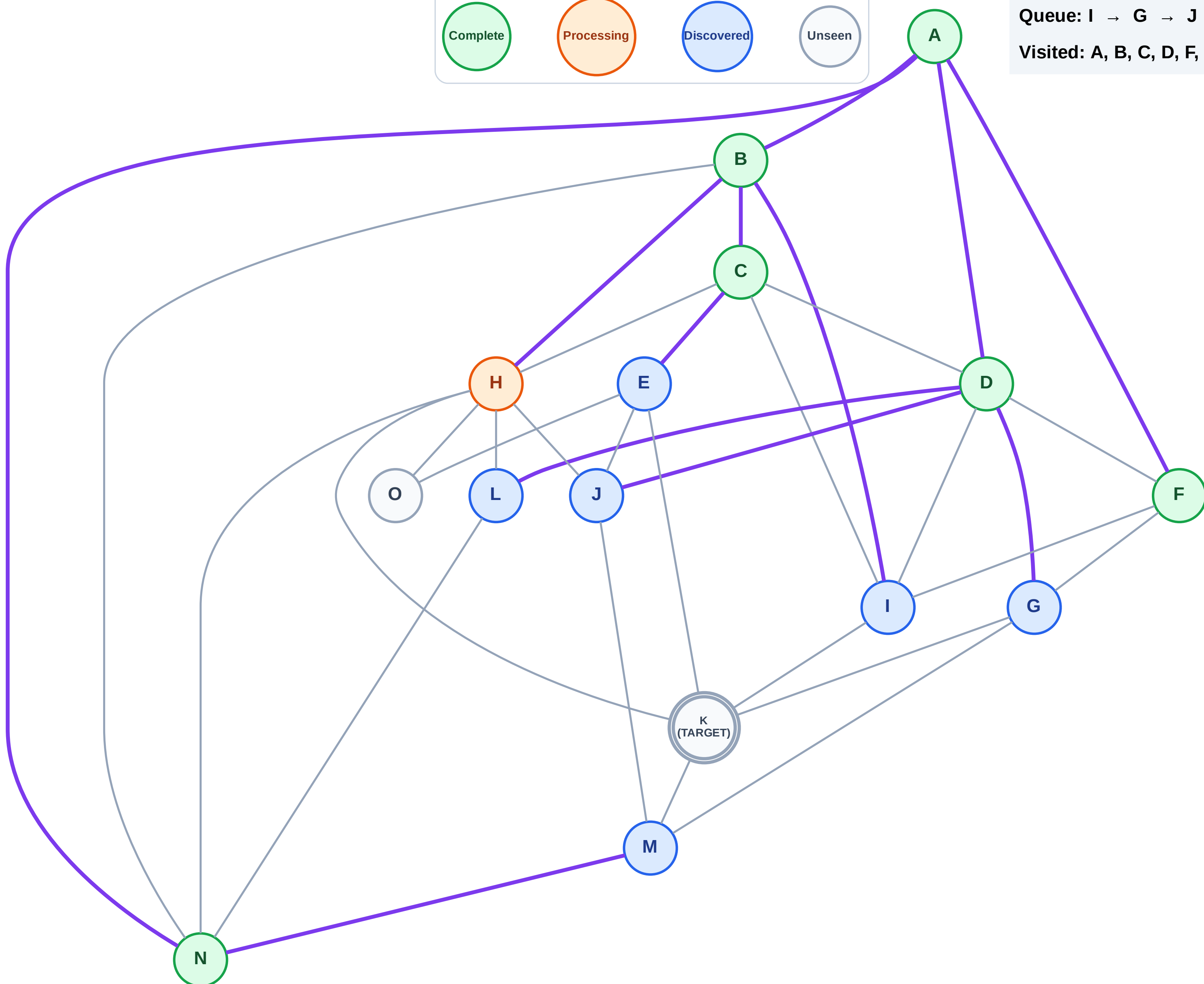
Discovered



Unseen

Queue: I → G → J → L → M → E

Visited: A, B, C, D, F, N



BFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → H → K

Legend

Complete

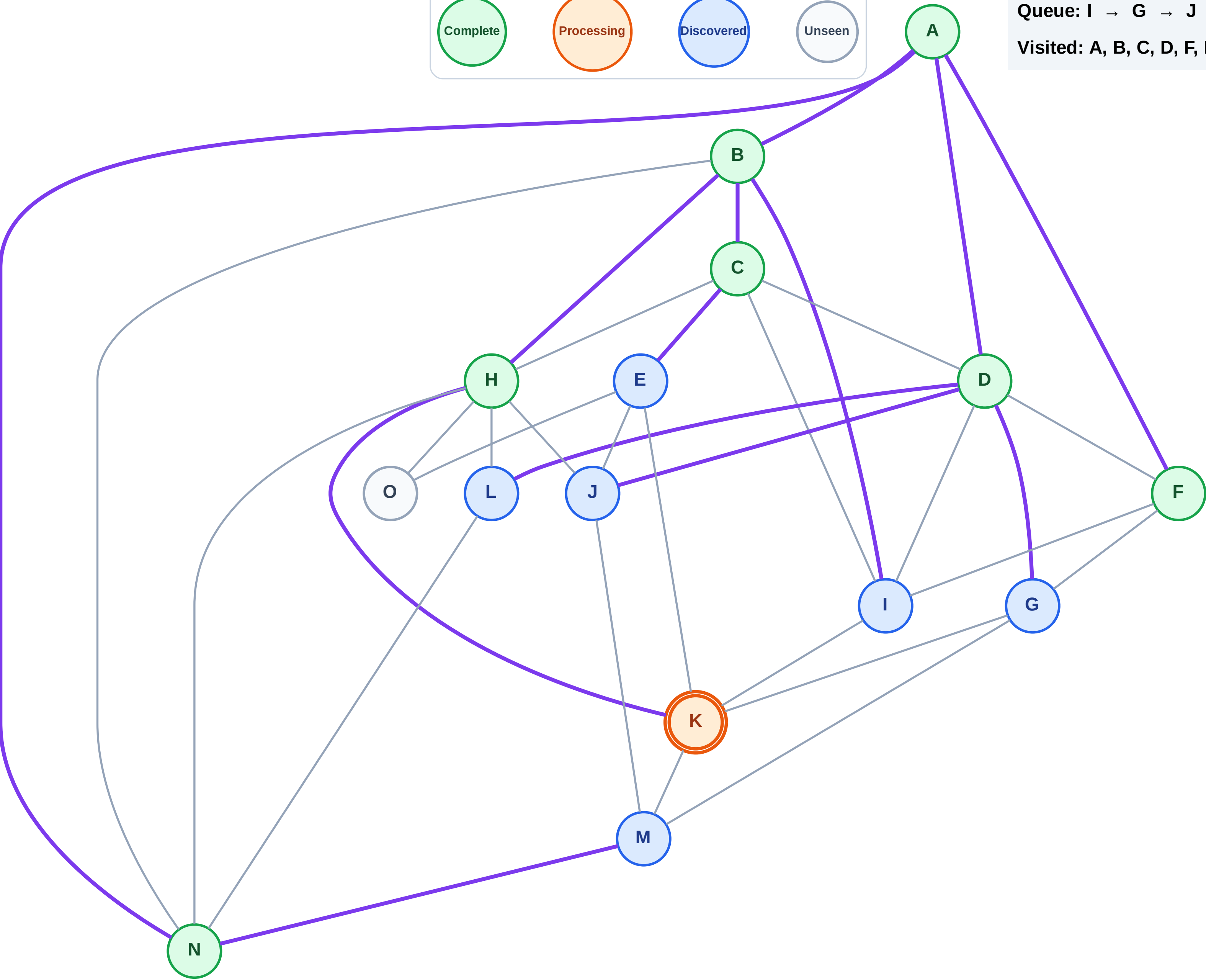
Processing

Discovered

Unseen

Queue: I → G → J → L → M → E

Visited: A, B, C, D, F, H, K, N



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

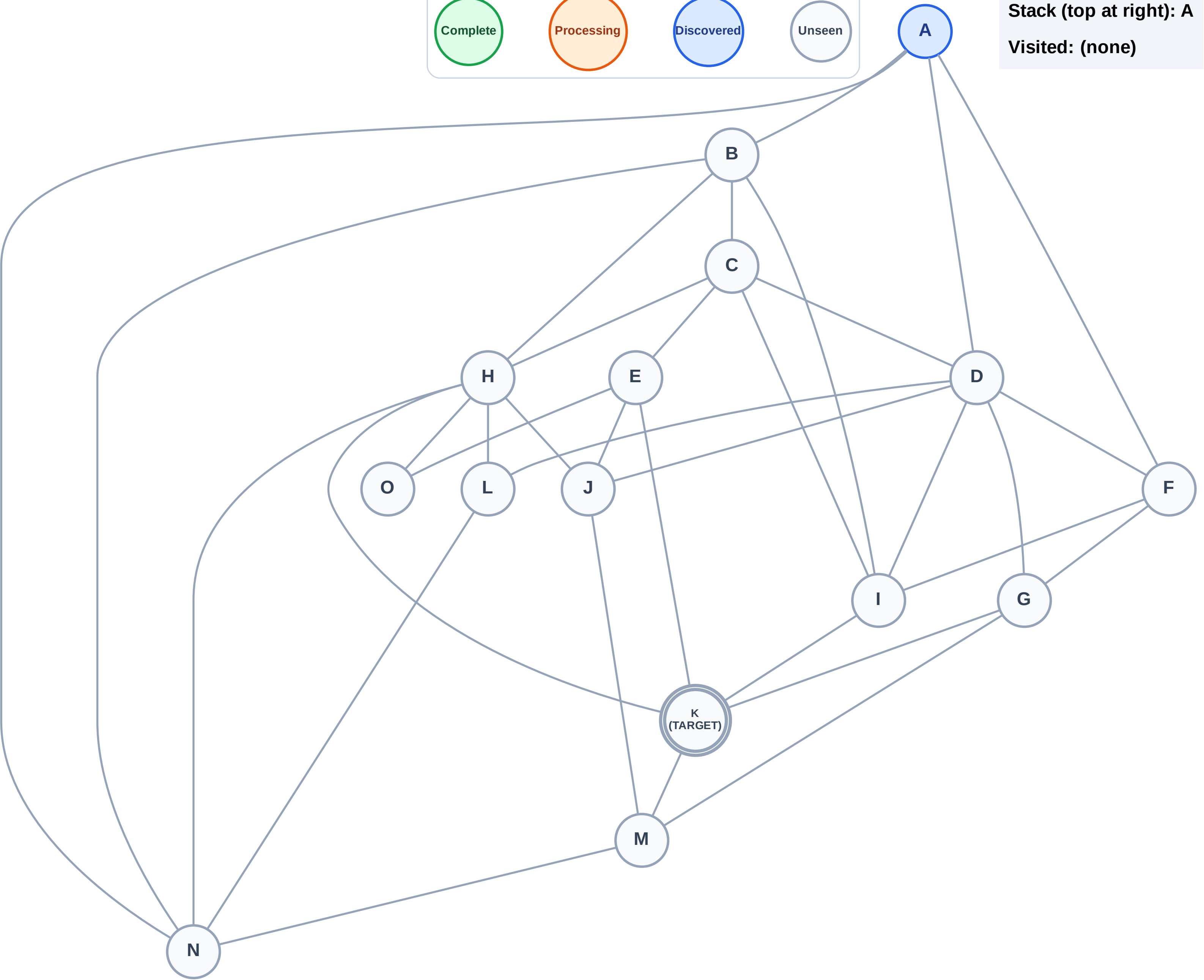
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

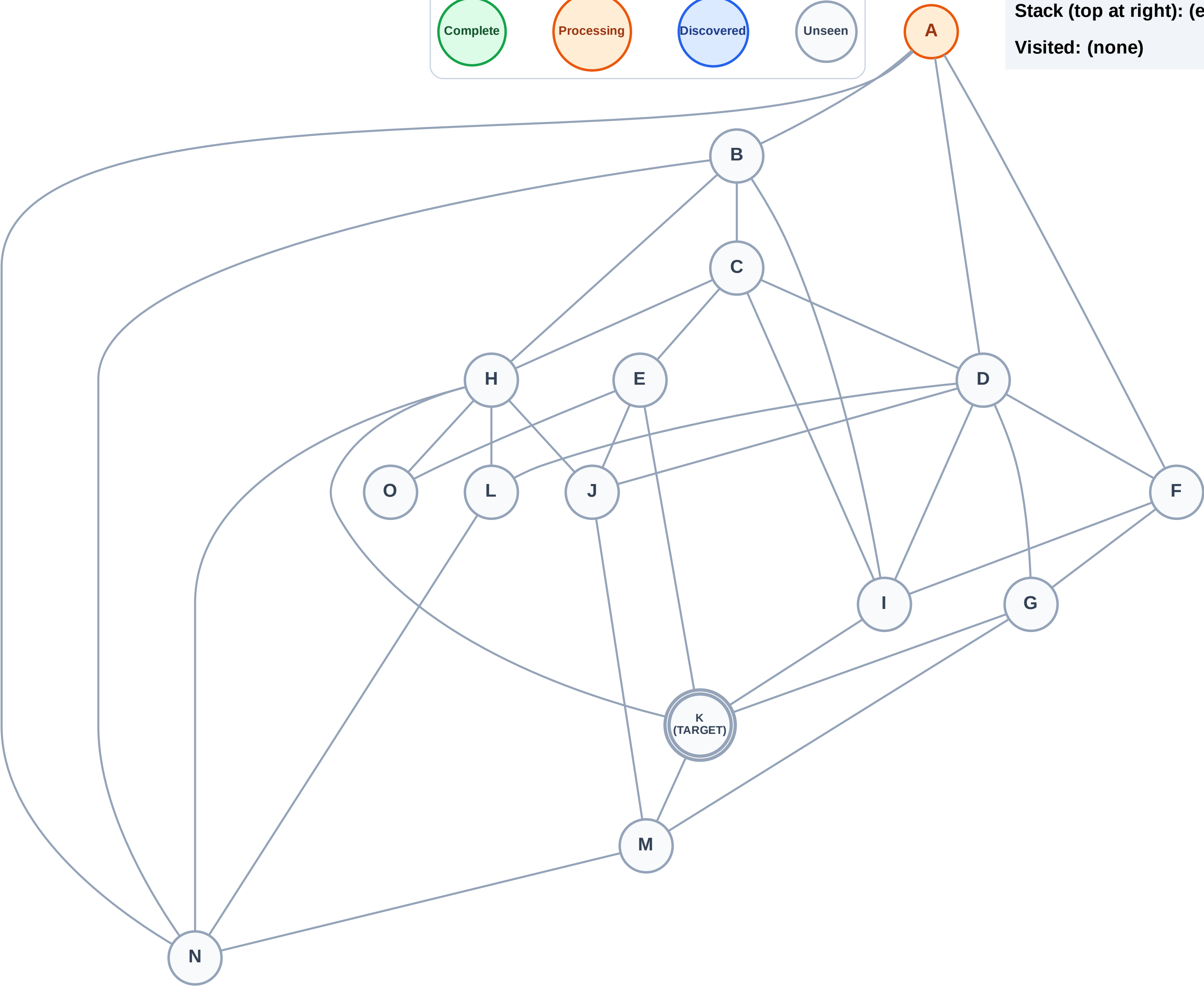
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

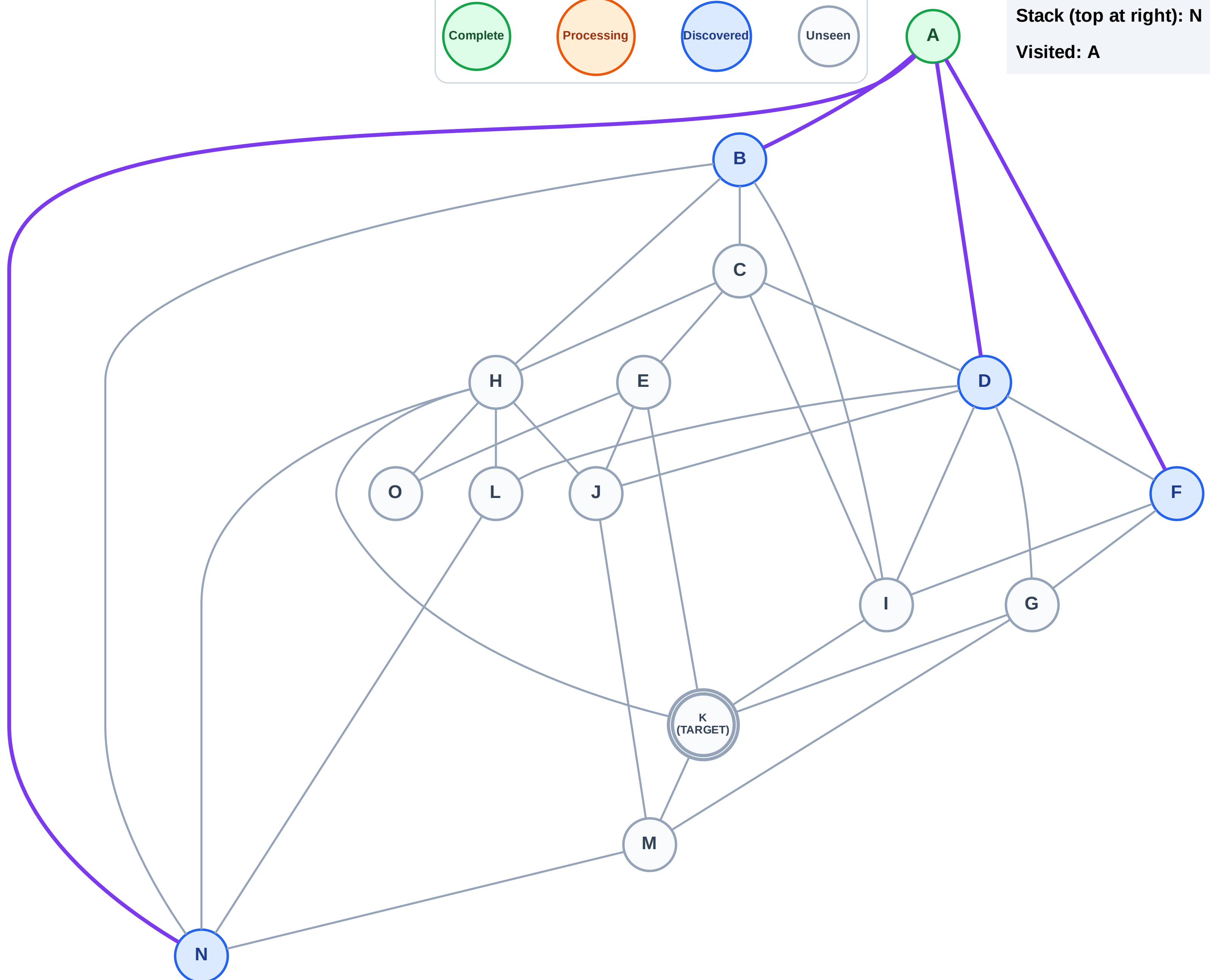
Step 3: Discover N, F, D, B from A

Legend



Stack (top at right): N | F | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

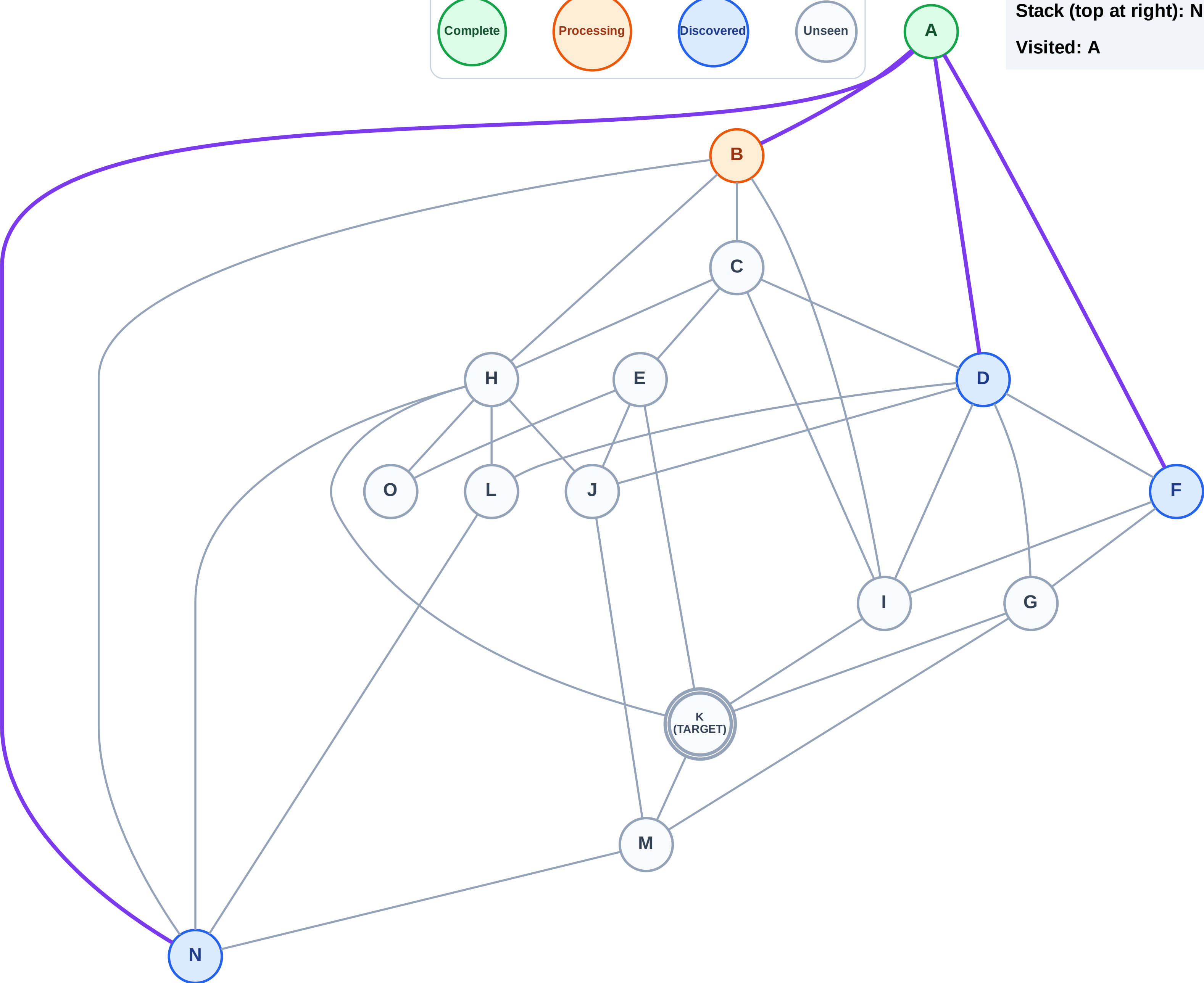
Processing

Discovered

Unseen

Stack (top at right): N | F | D

Visited: A



DFS Traversal

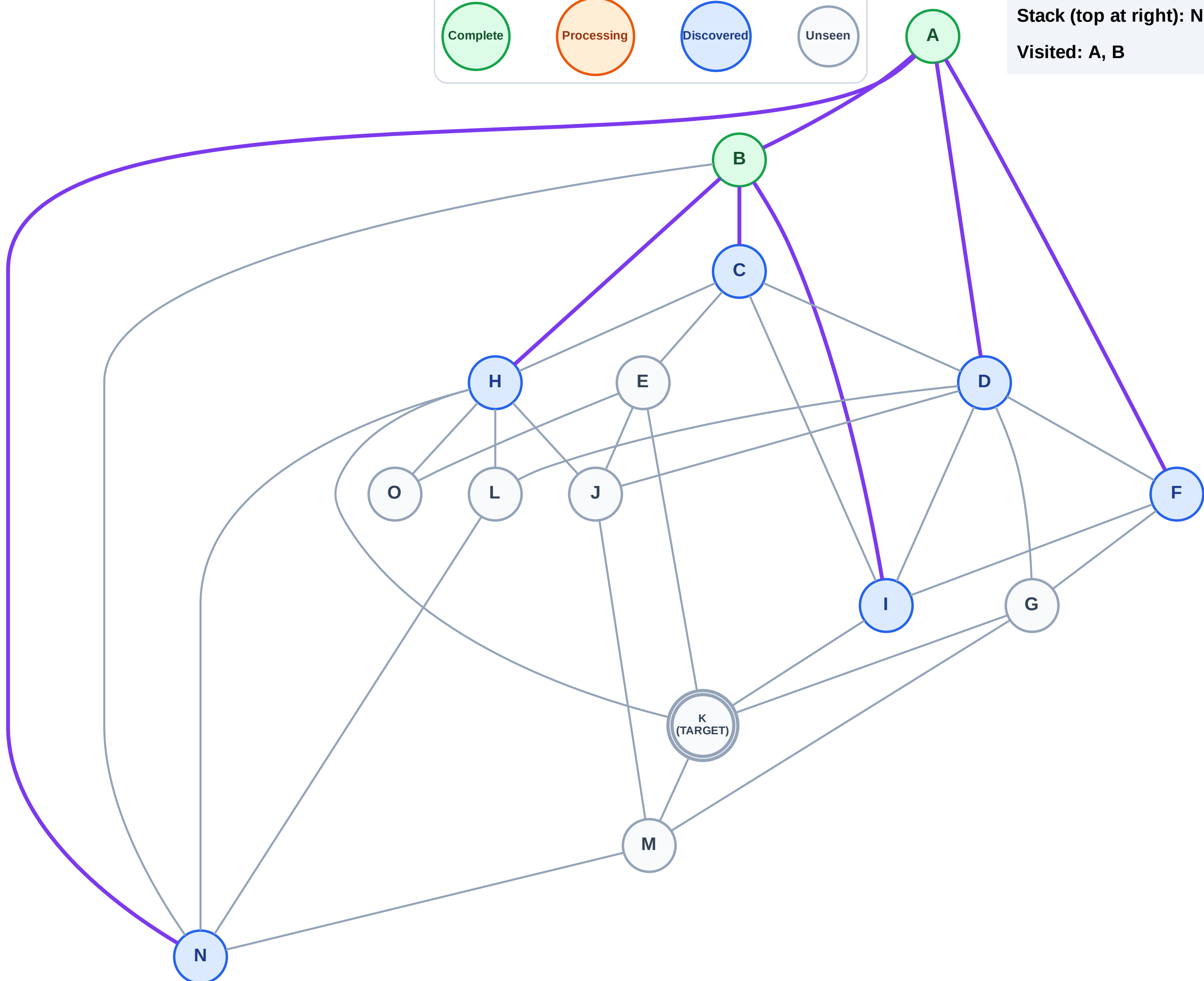
Step 5: Discover I, H, C from B

Legend



Stack (top at right): N | F | D | I | H | C

Visited: A, B



Step 6: Process node C

Step 6: Process node C

Complete

Complete



Processing

Processing



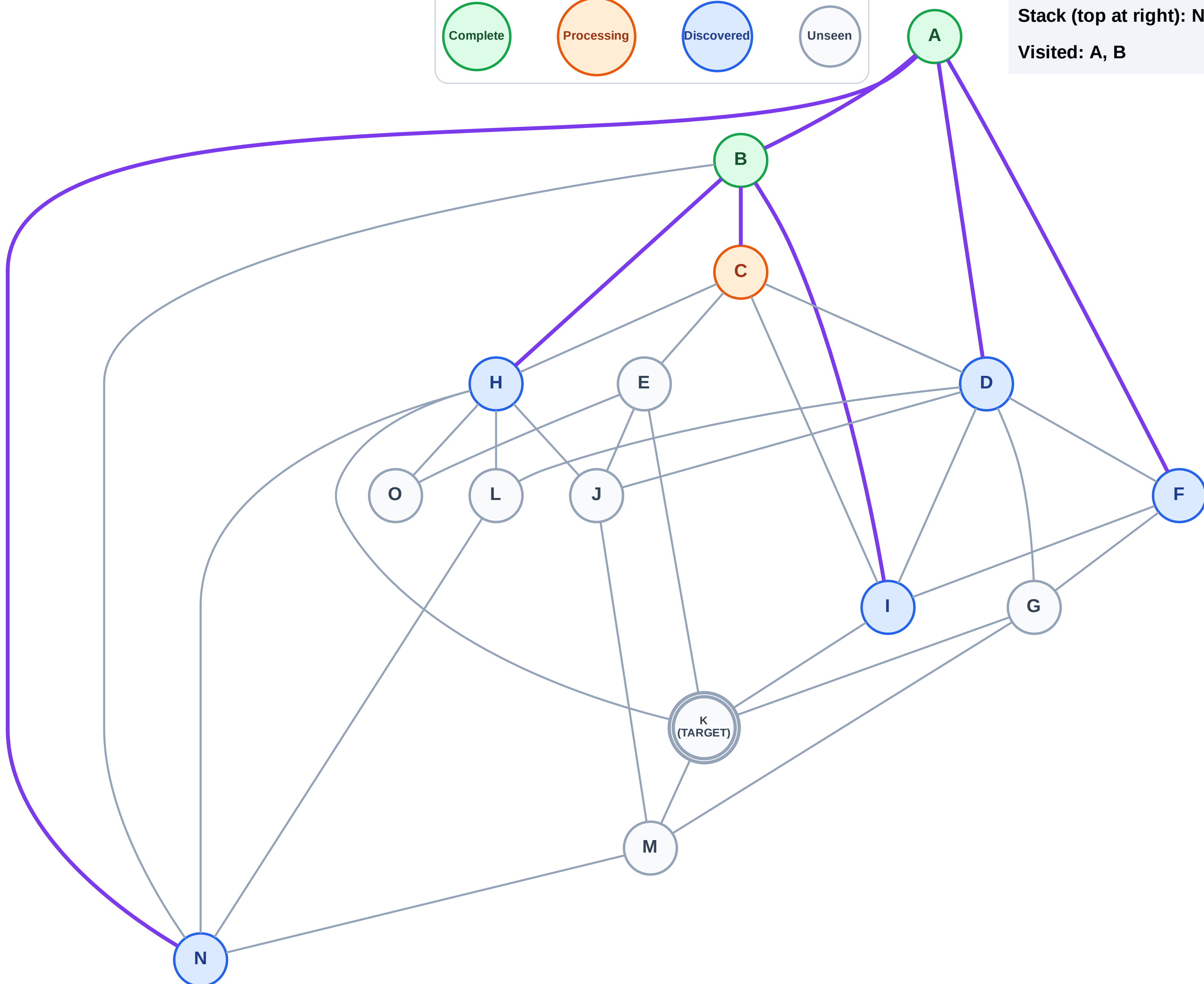
Discovered

Discovered



Unseen

Visited: A, B



DFS Traversal

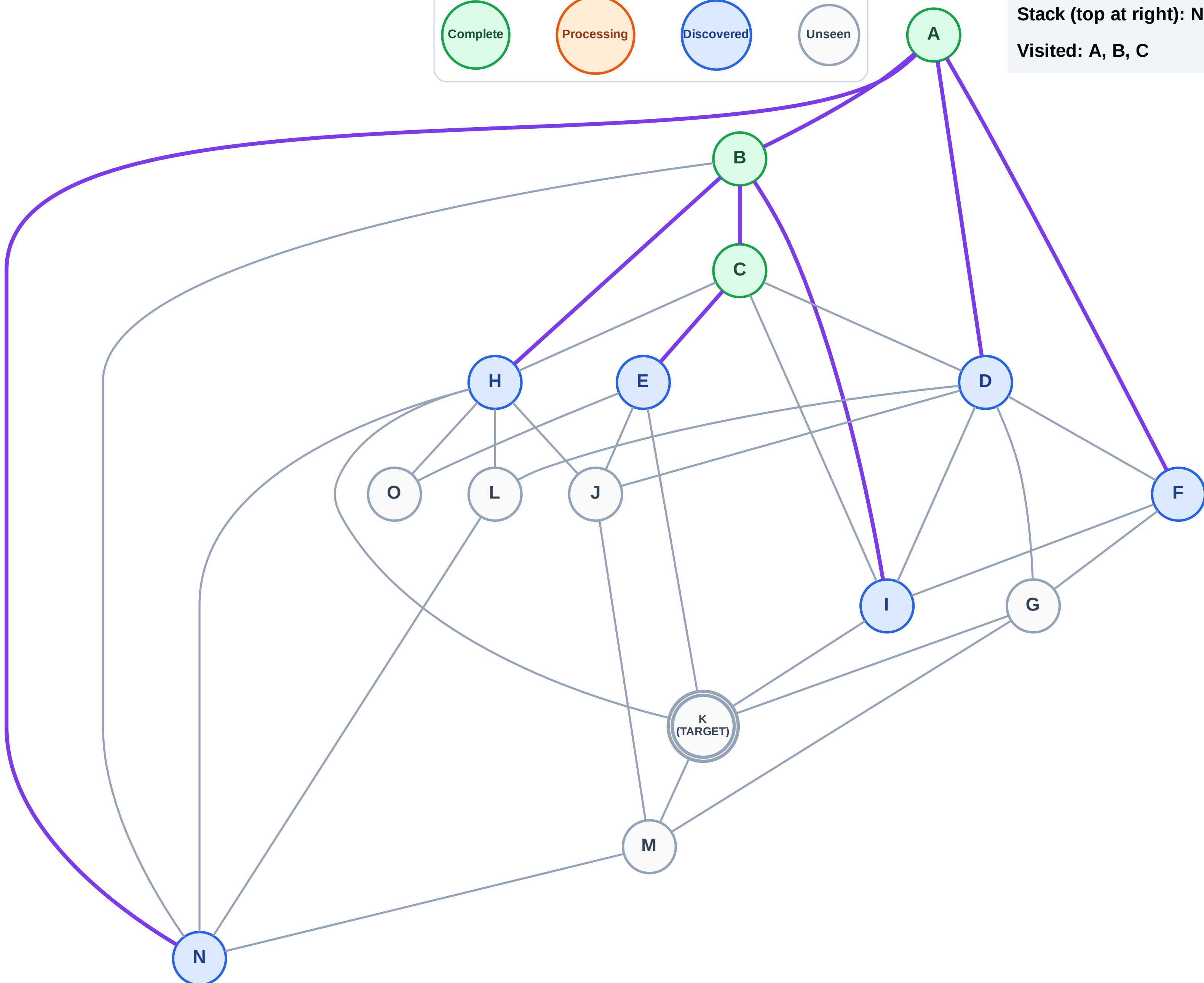
Step 7: Discover E from C

Legend



Stack (top at right): N | F | D | I | H | E

Visited: A, B, C



DFS Traversal

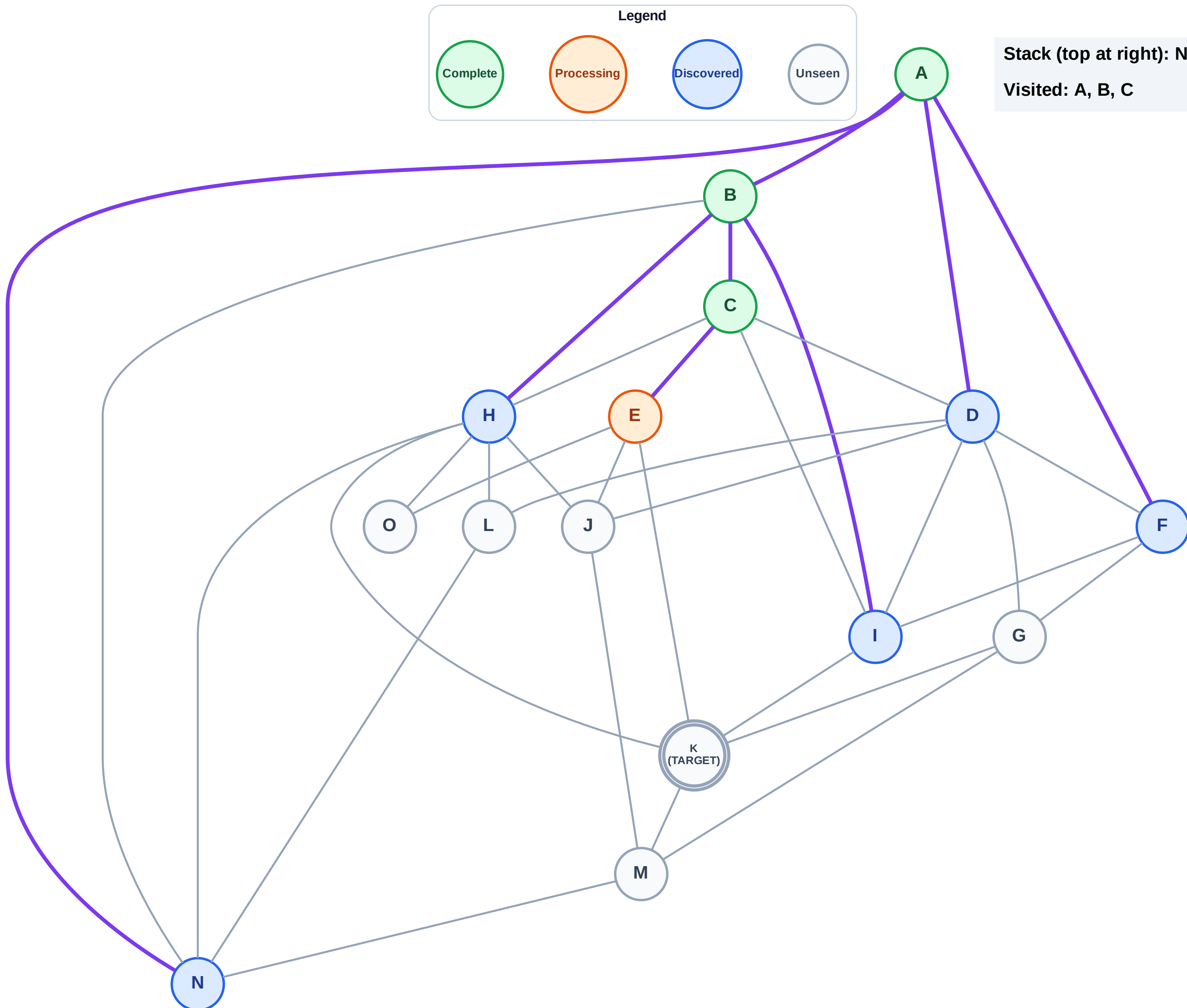
Step 8: Process node E

Legend



Stack (top at right): N | F | D | I | H

Visited: A, B, C



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | F | D | I | H | O
Visited: A, B, C, E, K

